

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
Total size of the file = 200 GB
Download speed = 2 GB / minute
Time taken to download the first 60 % of the file = 60 % of 200 GB / 2 GB / minute = 0.6 hours or 36 minutes
Time taken for the restart = 20 minutes
Total time taken so far = 36 minutes + 20 minutes = 56 minutes
Remaining size of the file = 200 GB - 40 % of 200 GB = 120 GB
Time taken to download the remaining 120 GB = 120 GB / 2 GB / minute = 60 minutes
Total time taken to download the file = 56 minutes + 60 minutes = 116 minutes
</think><answer>116 minutes</answer>

